

a Dated law families on either side of the horizon

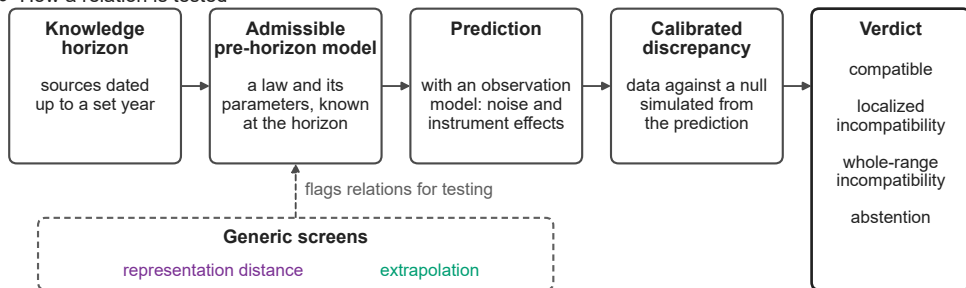
Pre-1900 law families with dated sources
(filled: training; open: held out)

Knowledge horizon: 1899

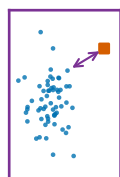
Later law families
(evaluation only)



b How a relation is tested

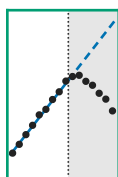


c Three kinds of evidence



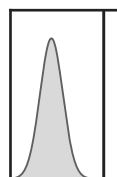
Representation screen

distance from pre-1900 relations in a learned embedding



Extrapolation screen

fit in the fitted regime (left); score the extrapolation region (shaded)

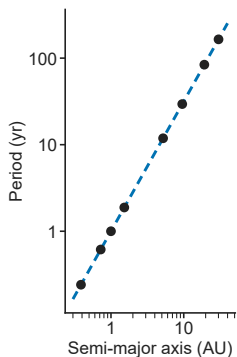


Named-prediction test

observed statistic (line) against a null simulated from the pre-1900 prediction (grey)

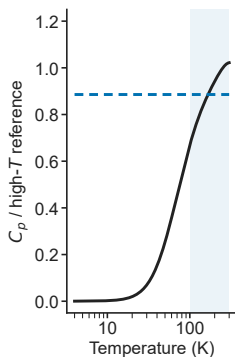
d Verdicts, shown on real data

Compatible



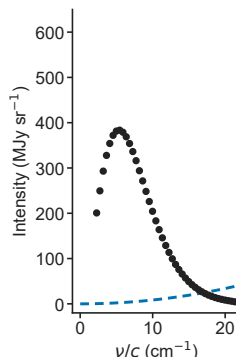
Solar-System orbits,
Kepler's law

Localized incompatibility



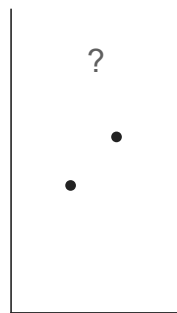
Copper heat capacity,
Dulong-Petit constant

Whole-range incompatibility



COBE-FIRAS spectrum,
Rayleigh-Jeans law

Abstention



too little evidence
(illustrative)